

Machine Learning for Natural Language Understanding

Prof. Achim Rettinger



Transformers

Chapter 7 – Decoding

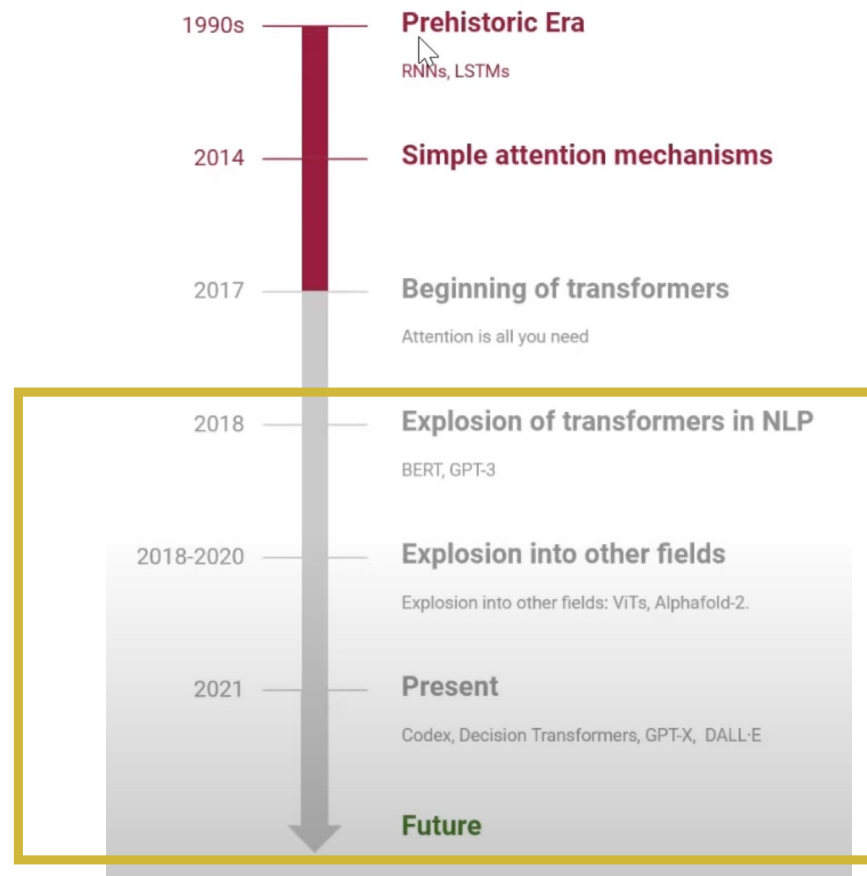
Sources: Lecture

- GPT3
 - <https://youtu.be/qGkzHFllWDY>
- Llama 3
 - <https://ai.meta.com/research/publications/the-llama-3-herd-of-models/>
- Autoregressive Transformers (GPT2)
 - <https://jalammar.github.io/illustrated-gpt2/>
- In-Context Learning
 - <https://arxiv.org/pdf/2005.14165.pdf>

Flipped Classroom Questions

- What block is not used in the GPT decoder compared to the Bert decoder?
- What is in-context learning?
- Name an example where decoders tend to fail.

Historic timeline – Focus of lecture today



<https://youtu.be/P127jhj-8-Y>

Foundation Models

Chapter 7.0.

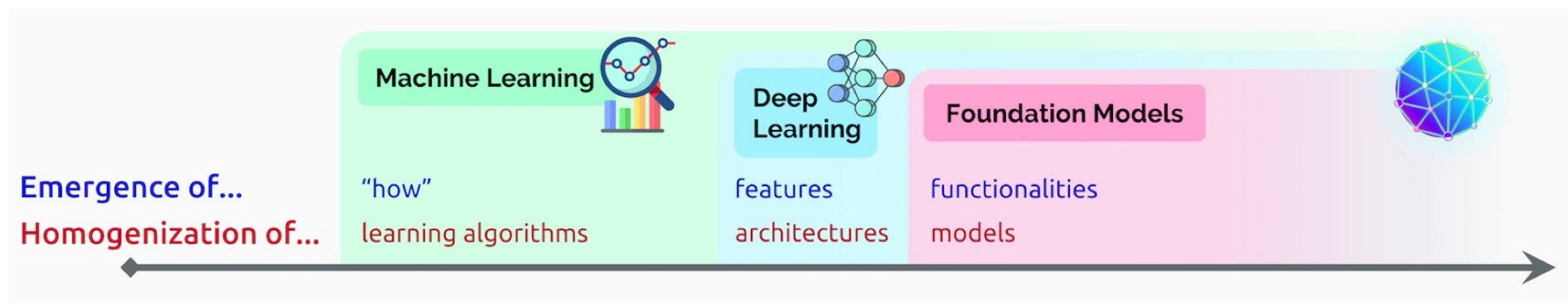
Definition “Foundation Models”

- A foundation model is any model that is trained on broad data (generally using self-supervision at scale) that can be easily adapted to a wide range of downstream tasks without extensive training.
- Most Large Language Models (LLMs) can be considered “Foundation Models”:
 - OpenAI's GPT models
 - Google's Gemini
 - Meta's LLaMa,
 - Anthropic's Claude
 - ...

https://en.wikipedia.org/wiki/Large_language_model#List_of_large_language_models

Emergence and Homogenization in AI

<https://arxiv.org/pdf/2108.07258.pdf>



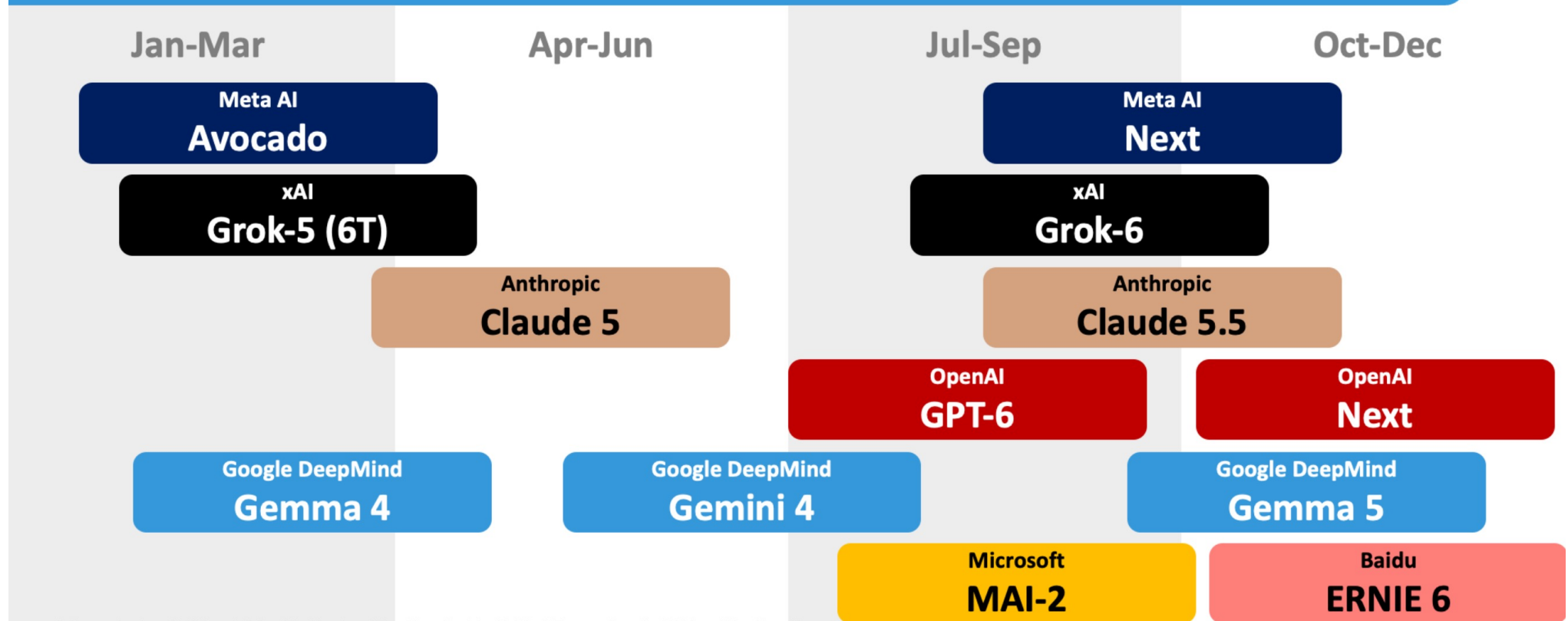
Transformer Decoder

Chapter 7.1



LLMs: NEXT PUBLIC RELEASES IN 2026

ESTIMATES AS OF
DECEMBER 2025



<https://lifearchitct.ai/timeline/>

Bert

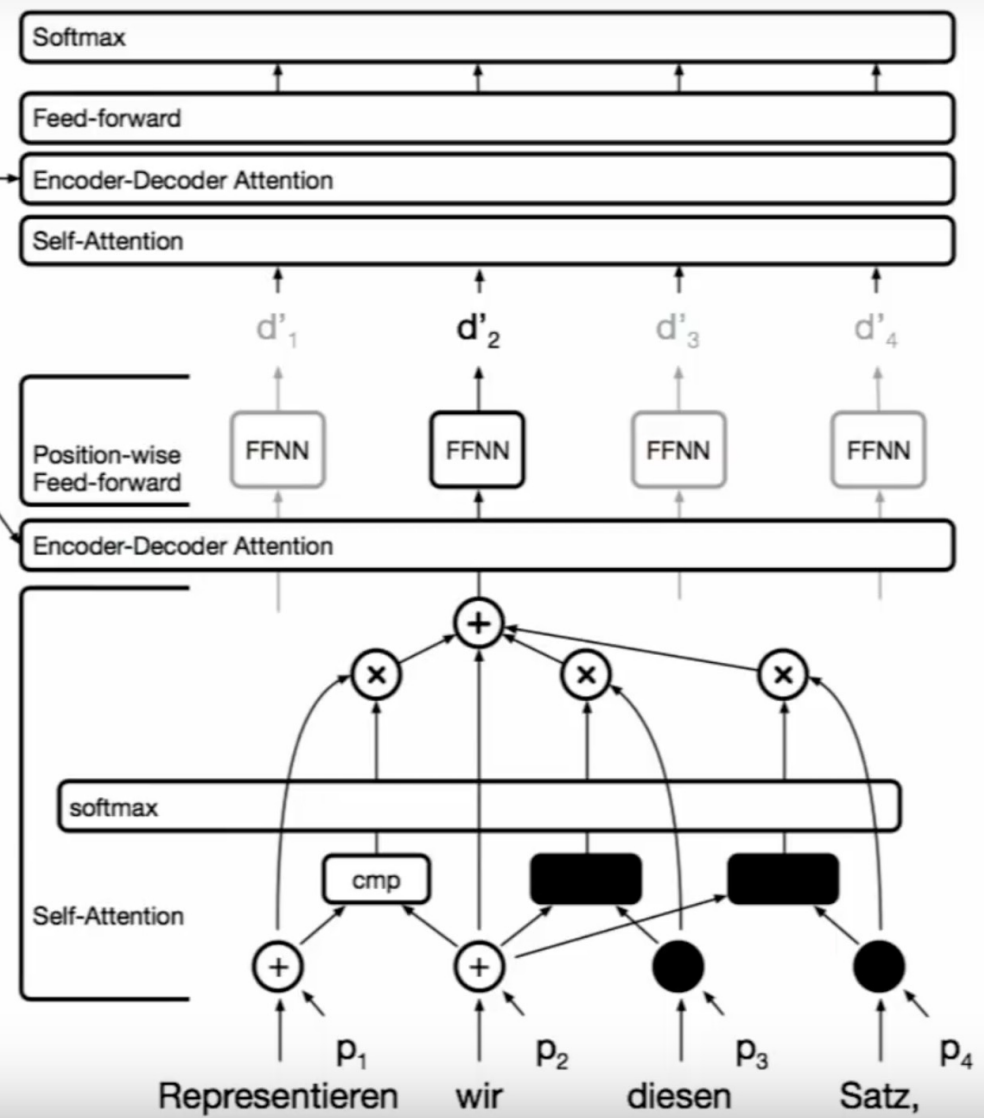
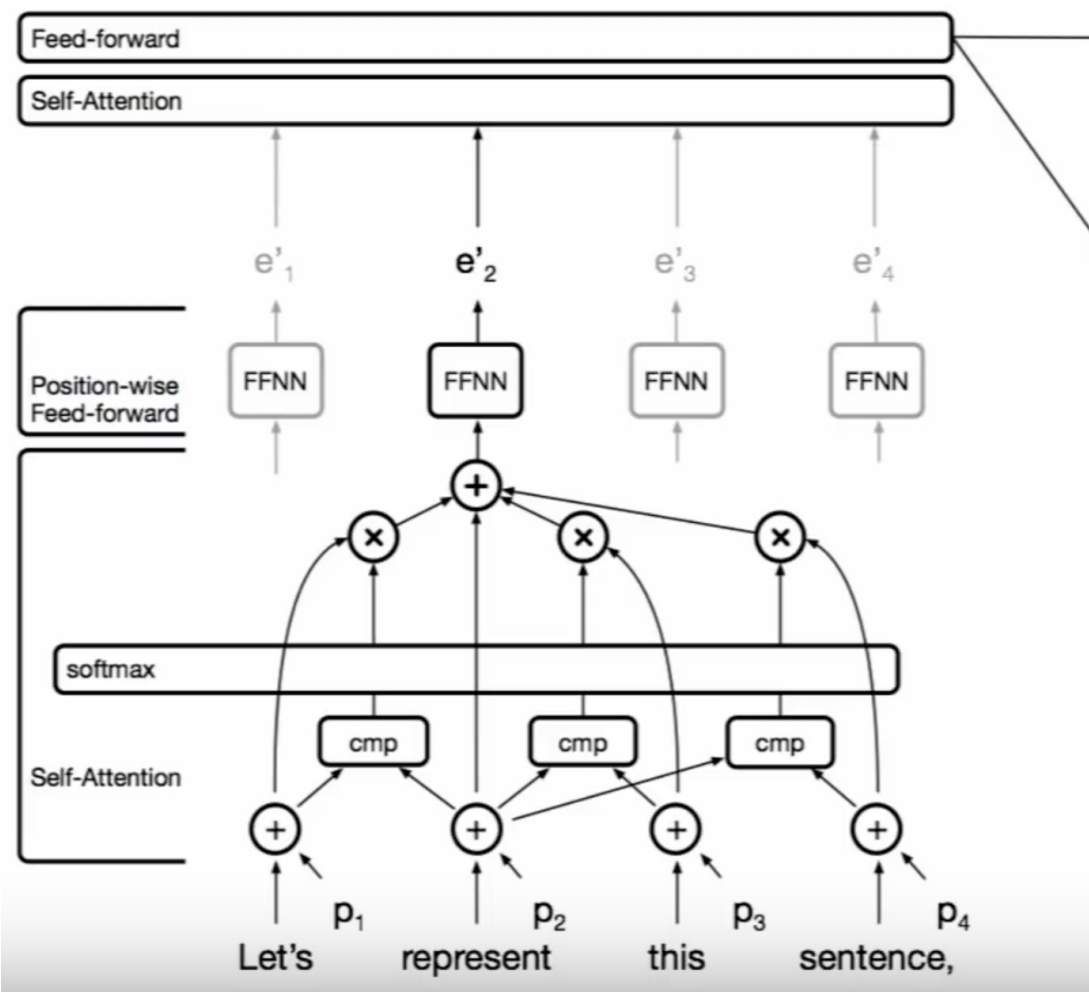
vs

GPT3

- Introduced 2018 by Google
- Bidirectional
- Primarily known for encoding (unsupervised pretraining + task-specific fine-tuning)
- Can be easily run on a single GPU (open-sourced)
- Google-Internally surpassed by PaLM
- DeepMind / Google

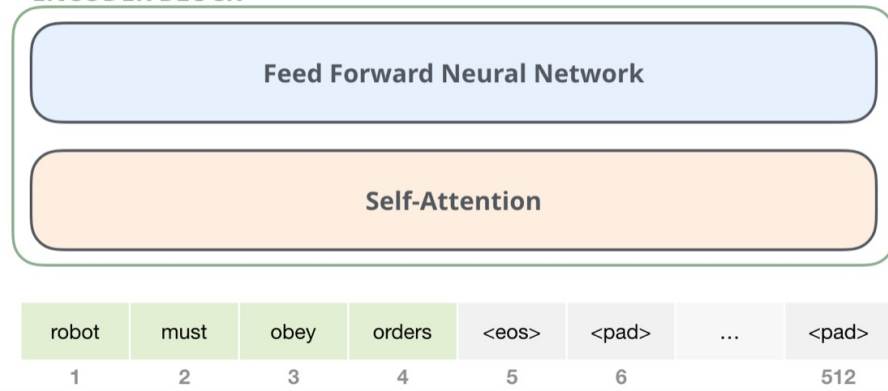
- Introduced 2020 by OpenAI
- Autoregressive (like LSTMs)
- Primarily used for decoding (prompting + language generation)
- Not easily run on commodity hardware (not open source)
- 470 times bigger than BERT-Large.
- OpenAI / Microsoft

Recap: Encoder-Decoder

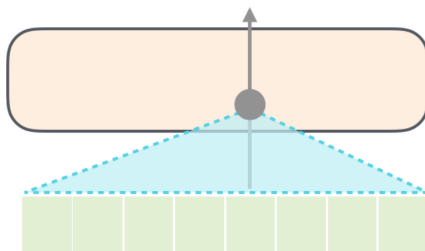


Encoder vs Decoder (Bert) (<https://jalammar.github.io/illustrated-gpt2/>)

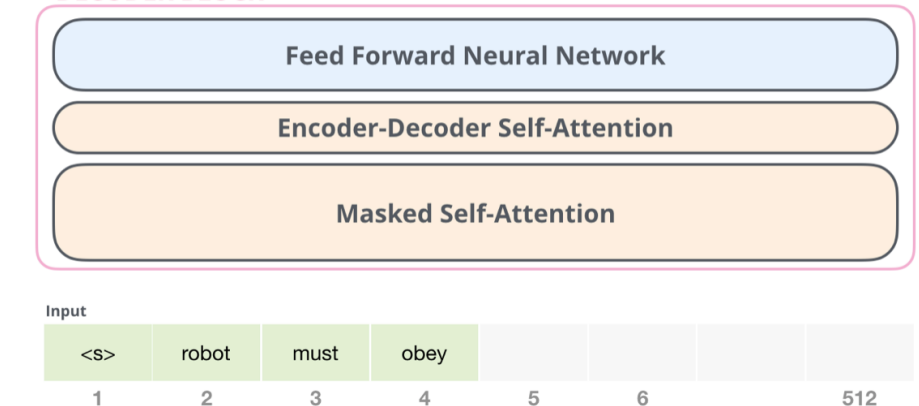
ENCODER BLOCK



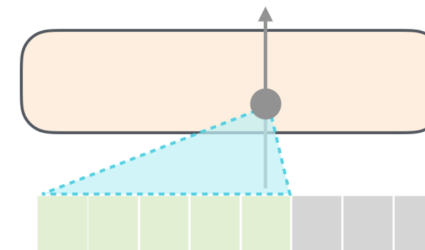
Self-Attention



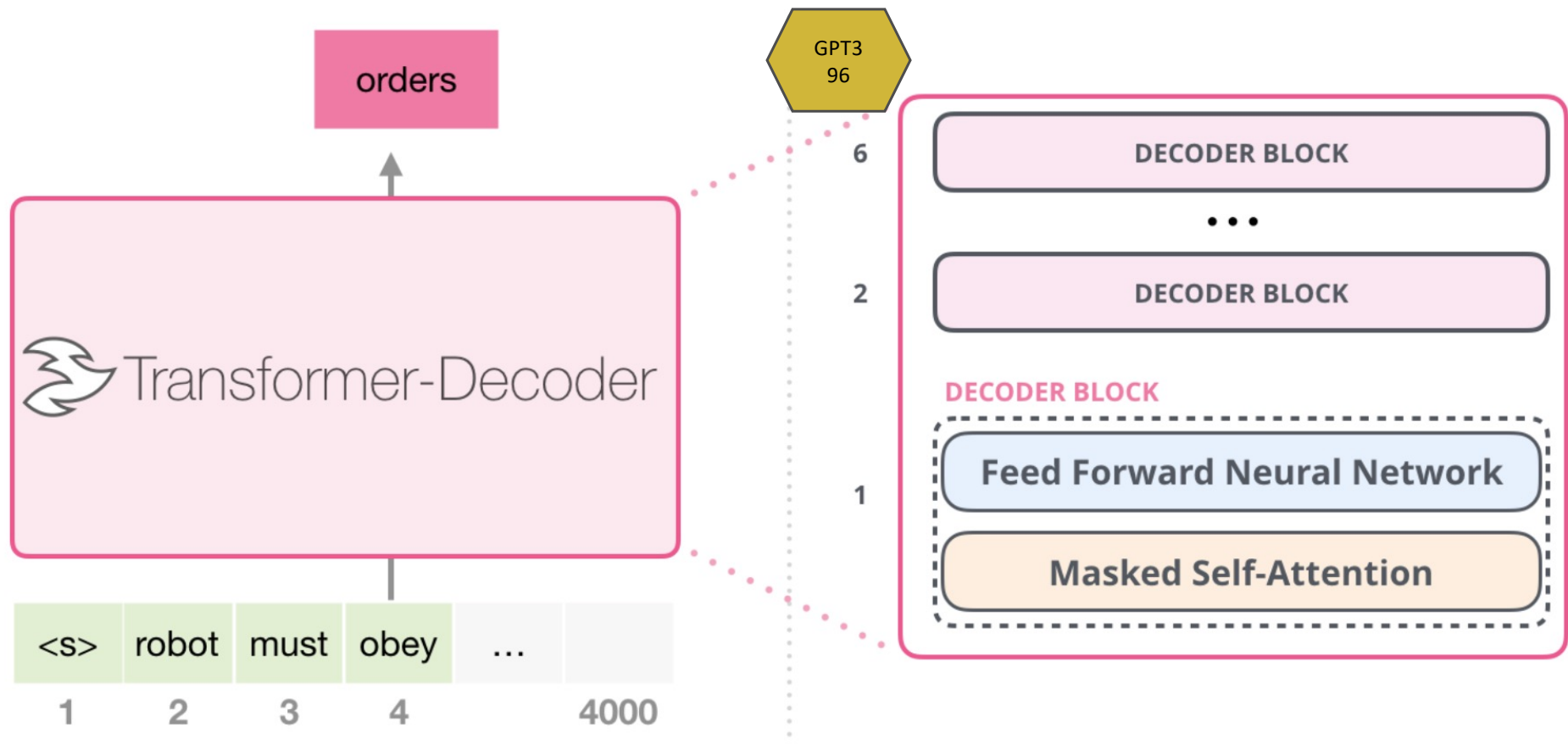
DECODER BLOCK



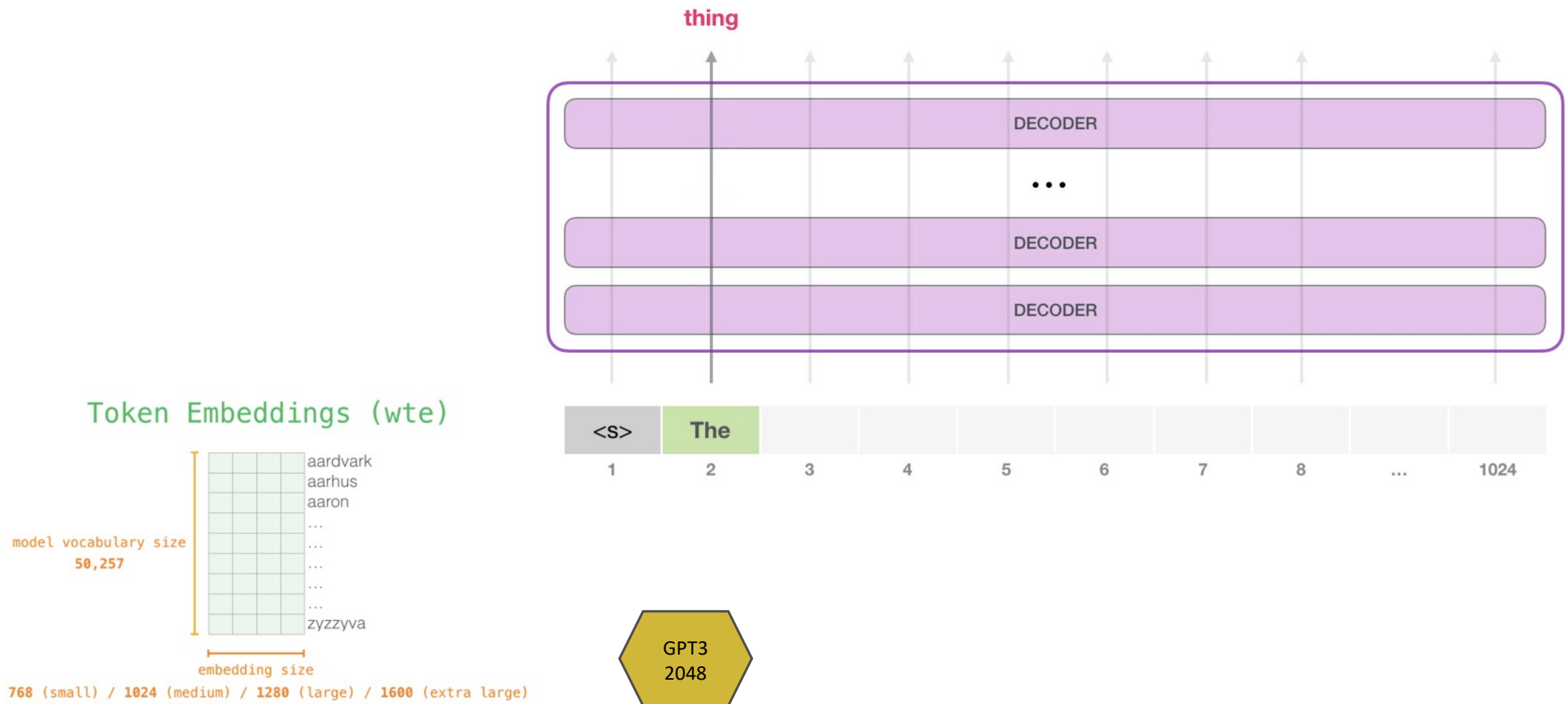
Masked Self-Attention



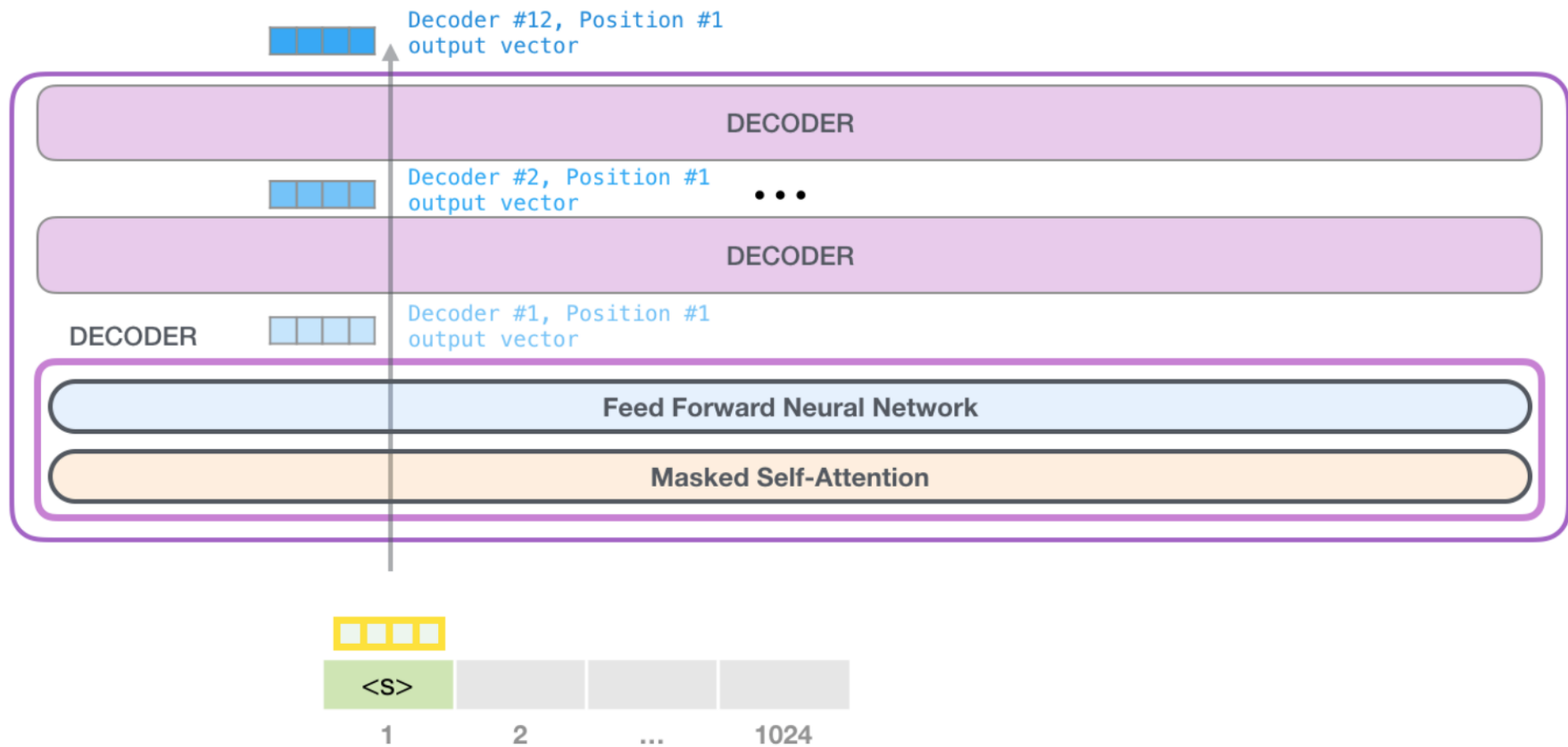
GPT uses Decoder-Only



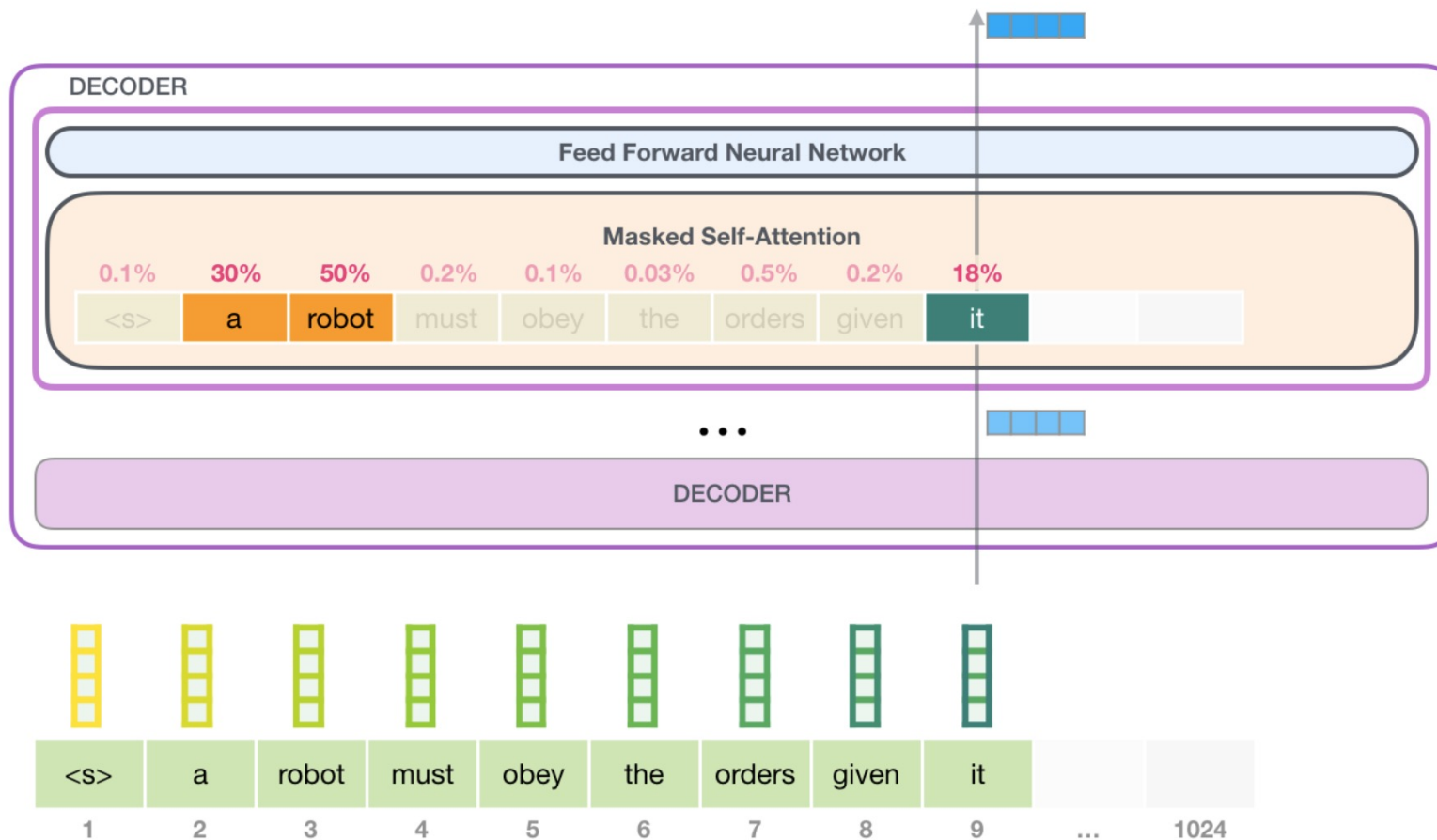
Autoregressive Generation



Processing the start token

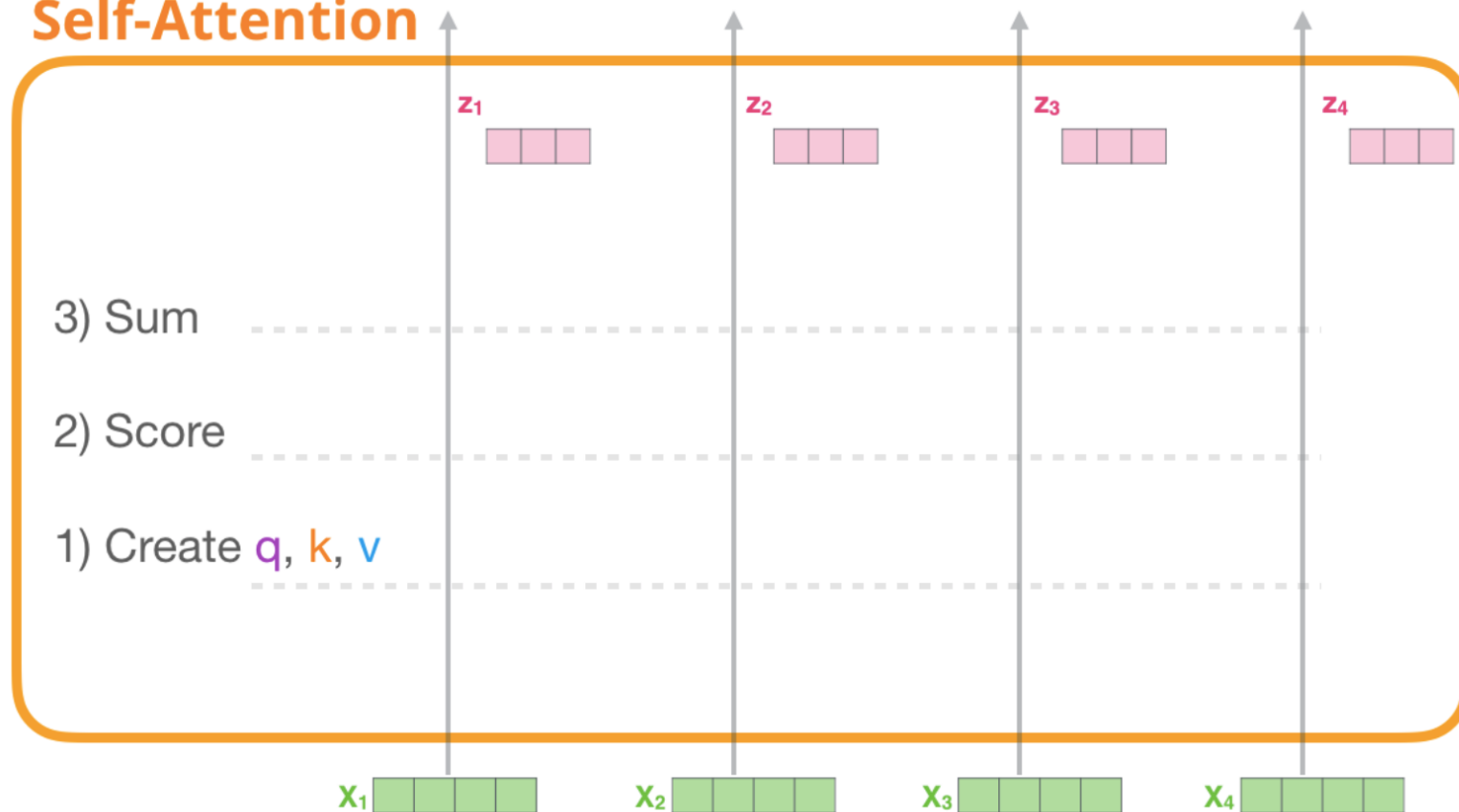


Self-Attention (unmasked)

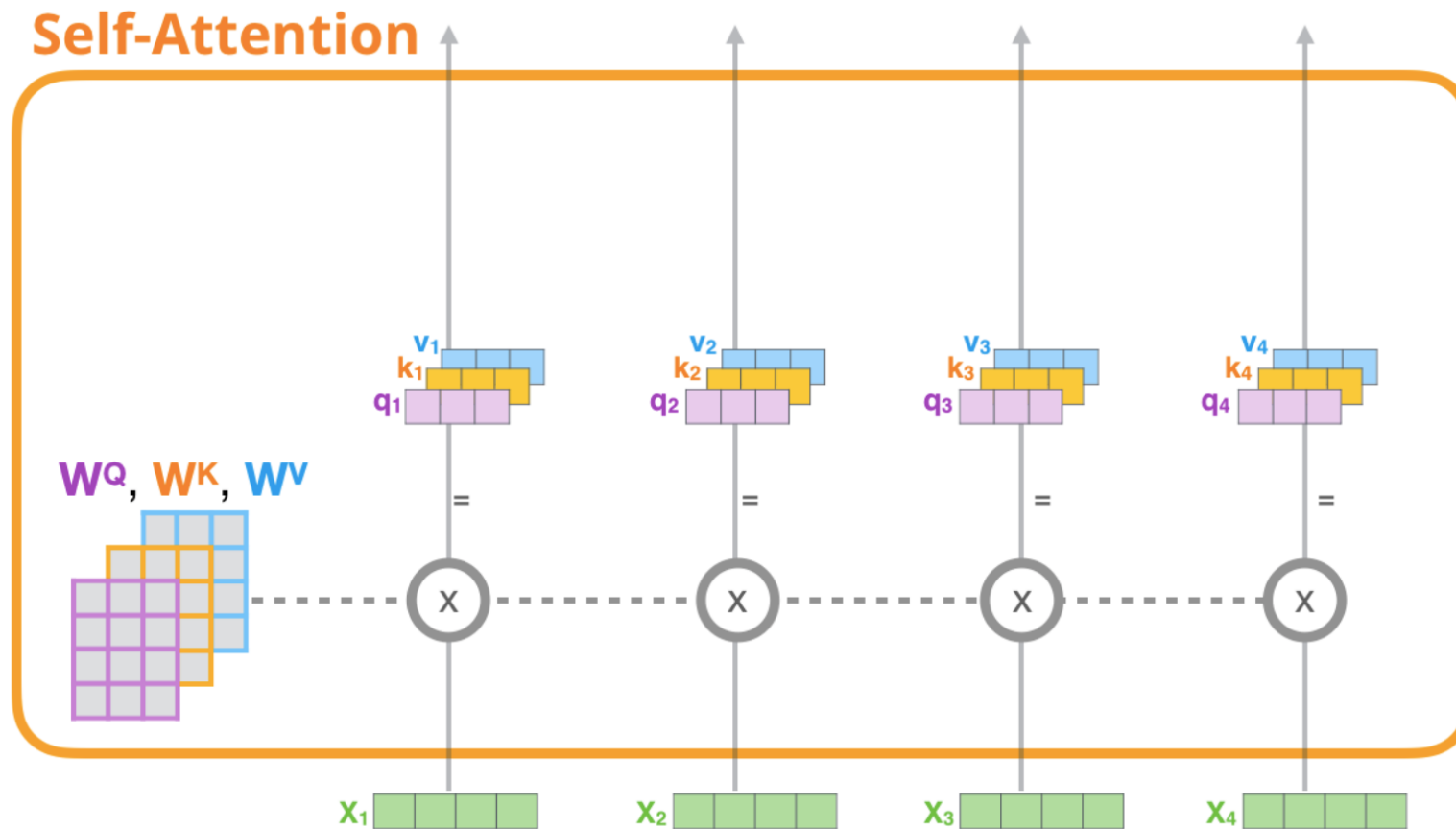


Create – Score – Sum

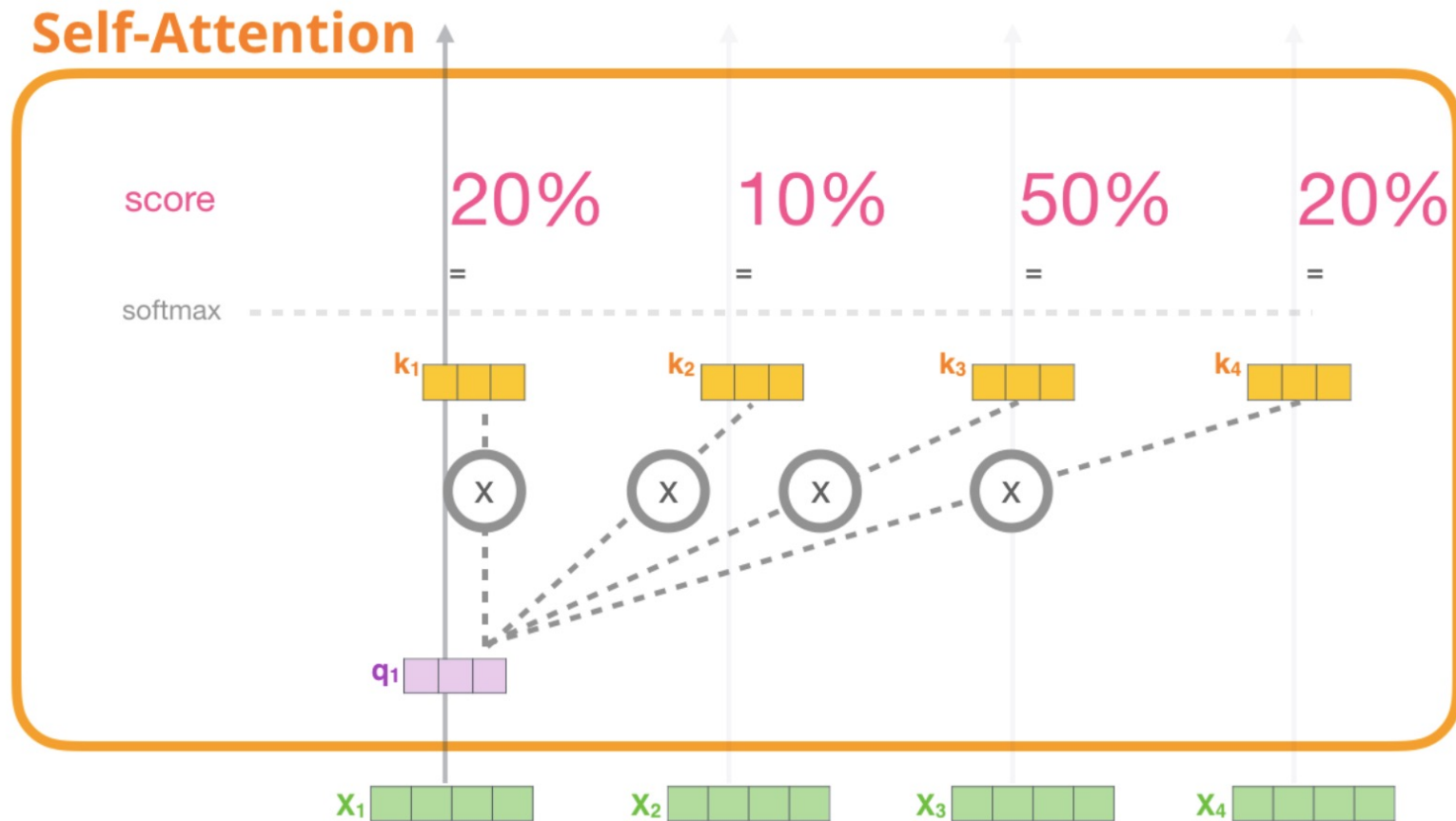
Self-Attention



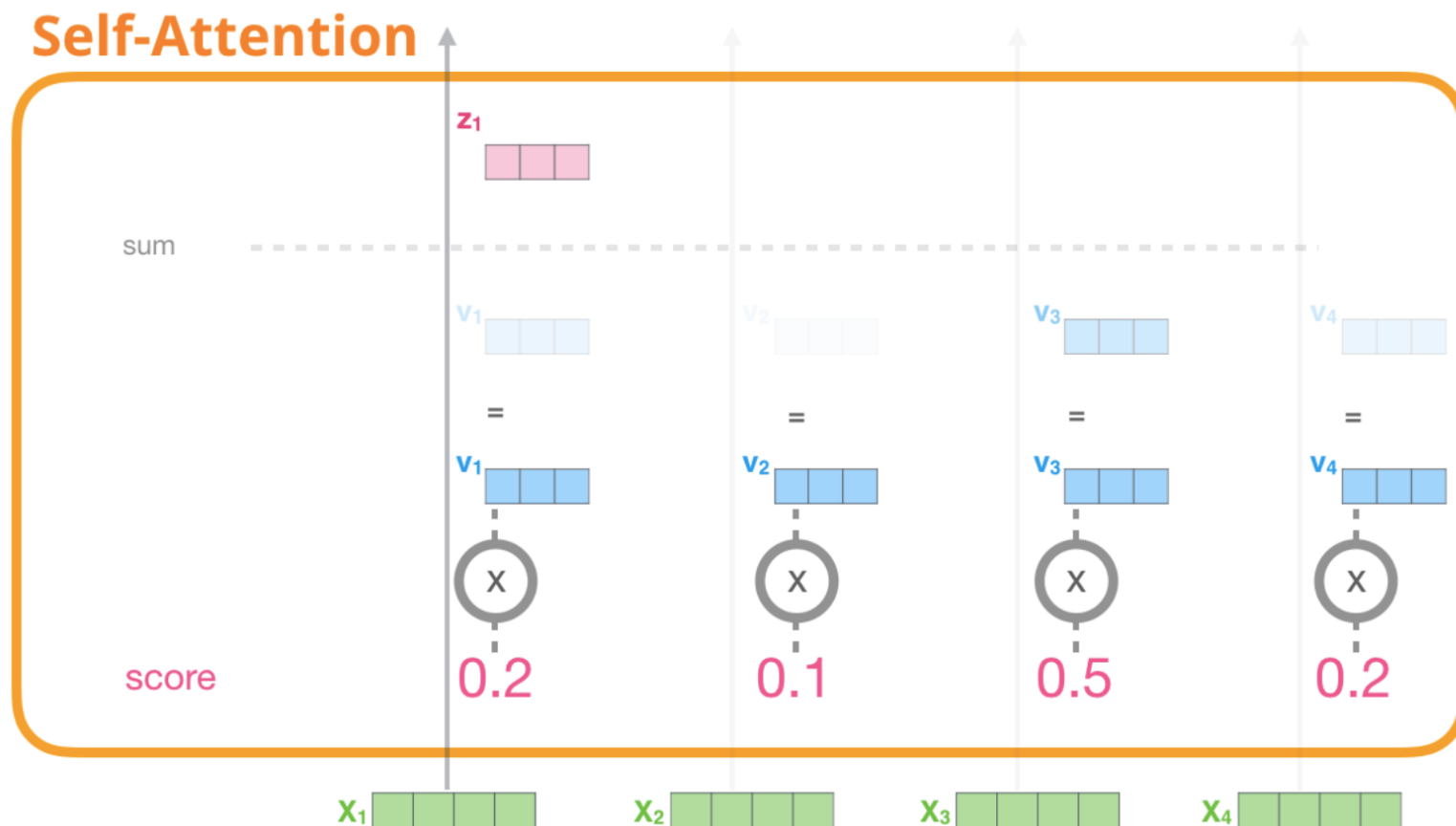
1) Create the Query, Key, and Value vectors



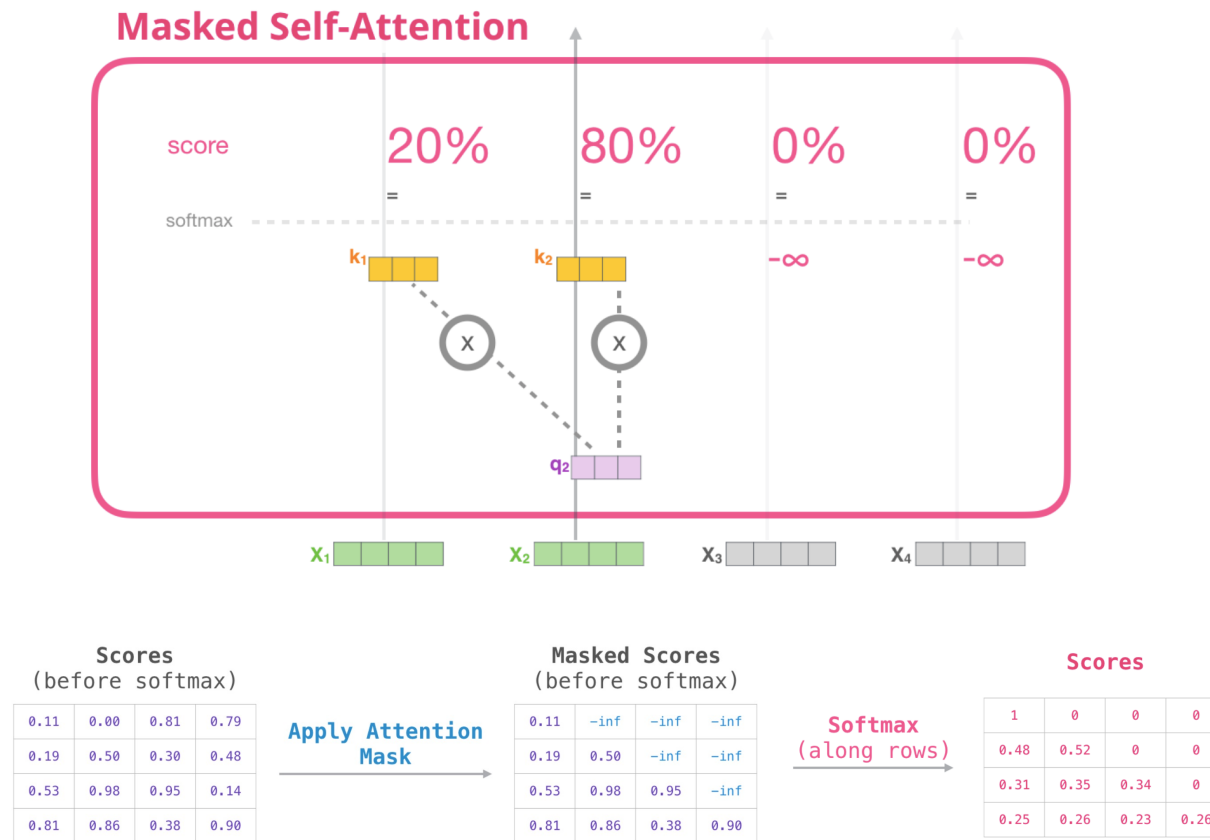
2) Multiply the query vector with each key vector



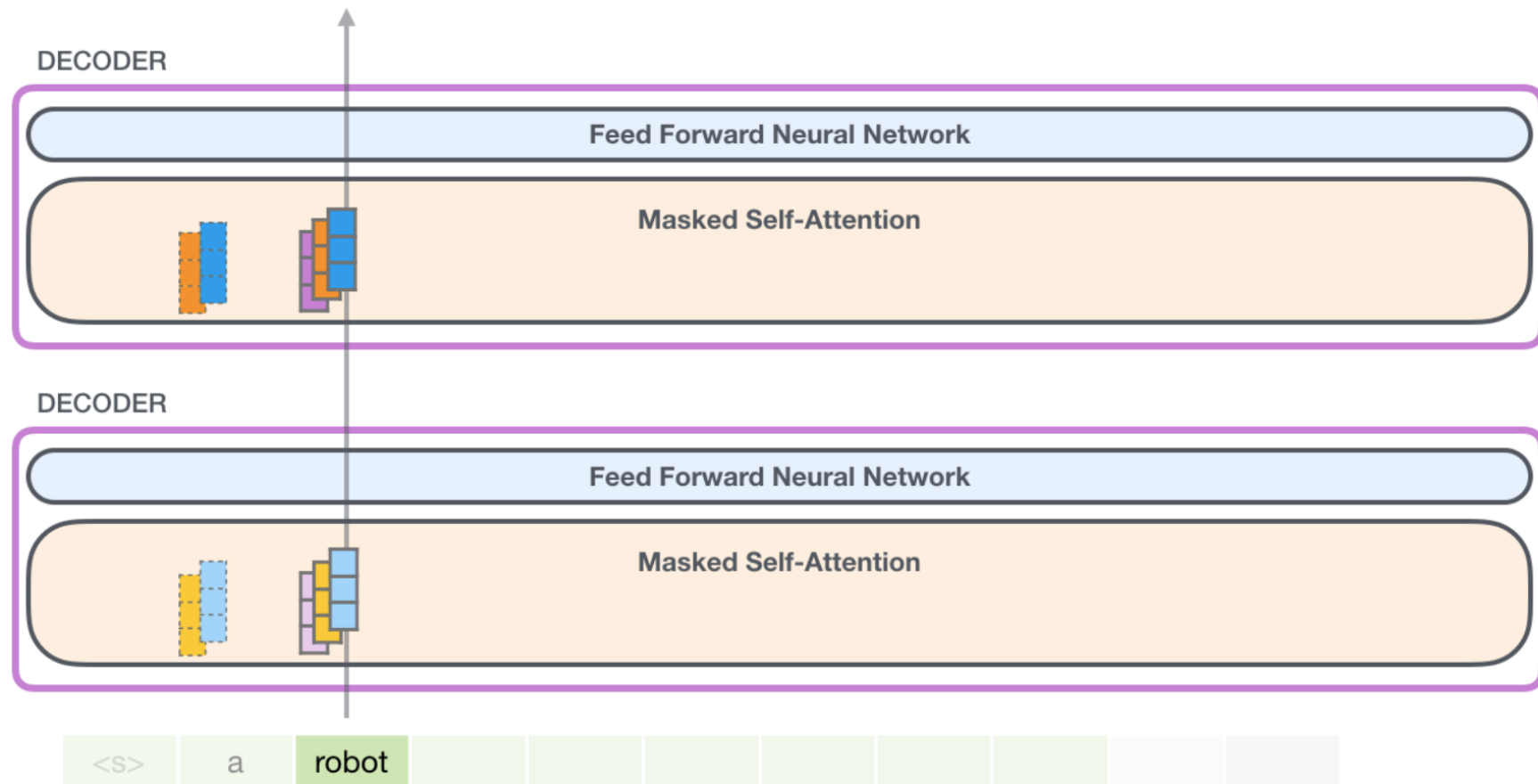
3) Weight and Sum the Value Vectors



GPT-2 Step 2) Masked Self Attention



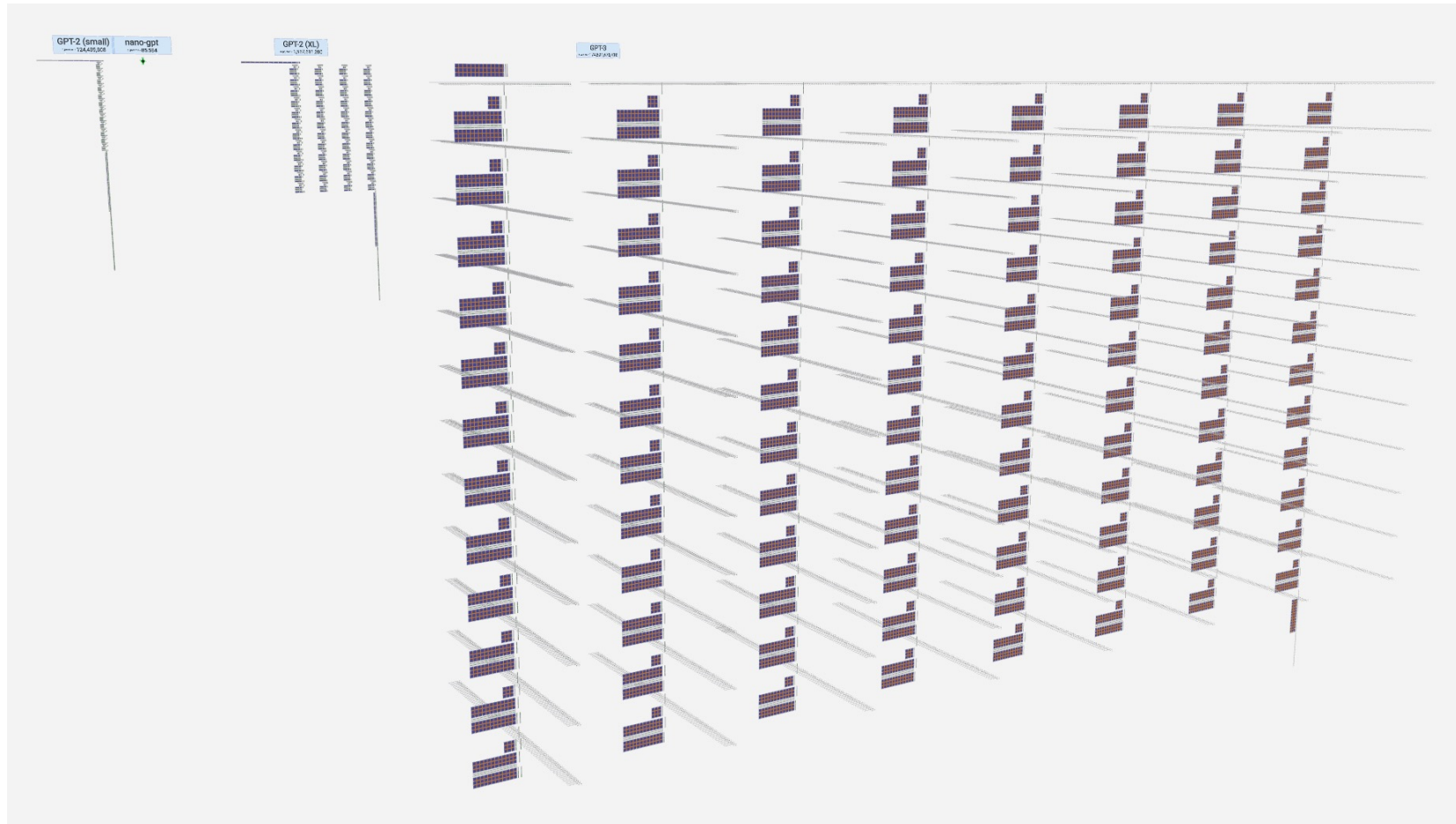
GPT-2 Masked Self-Attention



Key hyperparameters of Llama 3

	8B	70B	405B
Layers	32	80	126
Model Dimension	4,096	8192	16,384
FFN Dimension	14,336	28,672	53,248
Attention Heads	32	64	128
Key/Value Heads	8	8	8
Peak Learning Rate	3×10^{-4}	1.5×10^{-4}	8×10^{-5}
Activation Function	SwiGLU		
Vocabulary Size	128,000		
Positional Embeddings	RoPE ($\theta = 500,000$)		

Scaling LLMs <https://bbycroft.net/llm>



“The future of language is about the development of language models...”¹

1

Computerlinguistik | Universität Trier | cl.cldh.uni-trier.de